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Histogram Equalization Using OpenCV (Grayscale & Color Images)

Aim

To write a Python program using OpenCV to perform histogram equalization on both grayscale and color images to enhance image contrast and brightness.

The program performs the following operations:

- Read and display a grayscale image
- Plot histogram of the grayscale image
- Apply histogram equalization on grayscale image

- Read and display a color image
- Plot histogram of B, G, R channels
- Convert image to HSV color space
- Apply histogram equalization on the Value (V) channel
- Convert the enhanced image back to BGR format
- Display original and enhanced images with histograms

Software Used

- Anaconda – Python 3.7
- Jupyter Notebook / VS Code
- OpenCV (cv2)
- NumPy
- Matplotlib

Algorithm

Step 1:

Import the required libraries: OpenCV, NumPy, and Matplotlib.

Step 2:

Read the image `parrot.jpg` in grayscale format.

Step 3:

Display the grayscale image and plot its histogram.

Step 4:

Apply histogram equalization using `cv2.equalizeHist()` to enhance contrast.

Step 5:

Display original grayscale image, its histogram, enhanced image, and its histogram using a 2×2 grid.

Step 6:

Read the same image in color format.

Step 7:

Split the image into B, G, R channels and plot their histograms.

Step 8:

Convert the image from BGR to HSV color space.

Step 9:

Apply histogram equalization on the V (Value) channel.

Step 10:

Merge the channels and convert the image back to BGR format.

Step 11:

Display original color image, histogram, enhanced image, and enhanced histogram using a 2×2 grid.

Program

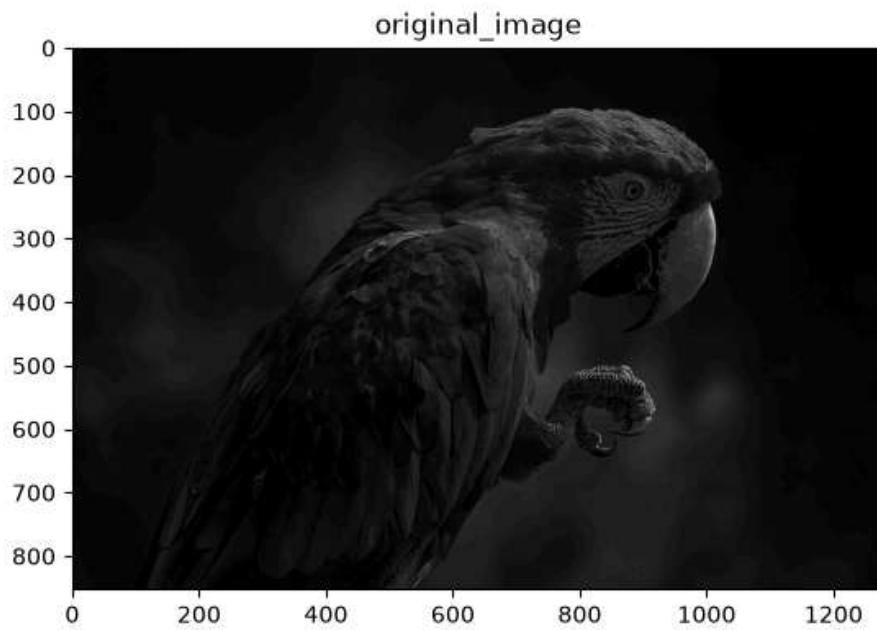
Developed By:

Name: Rakisha R

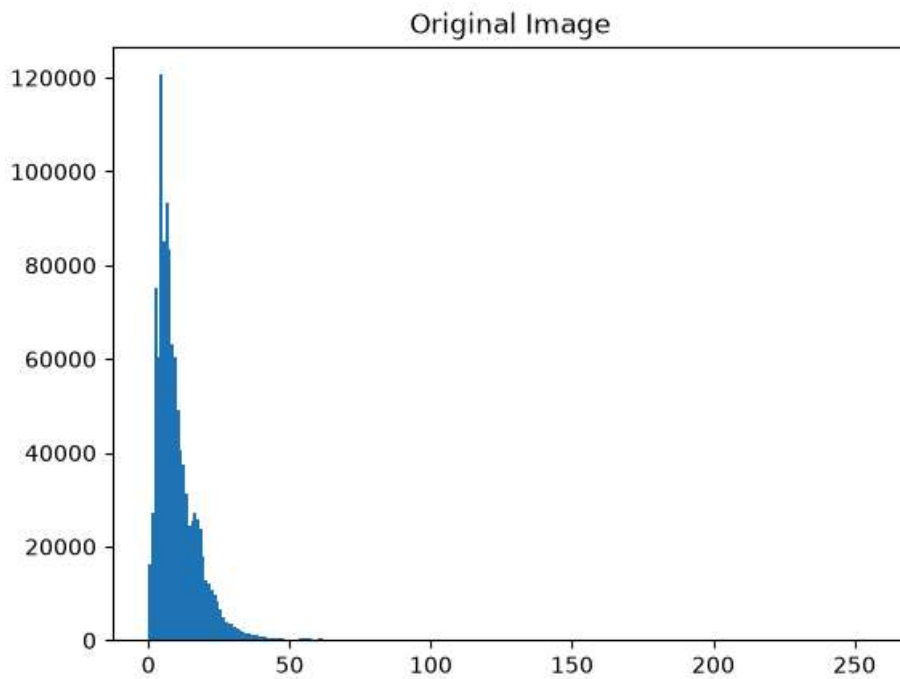
Register No: 212225230223

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
img = cv2.imread('parrot.jpg',cv2.IMREAD_GRAYSCALE)
plt.imshow(img, cmap='gray')
plt.title('original_image')
plt.show()
```





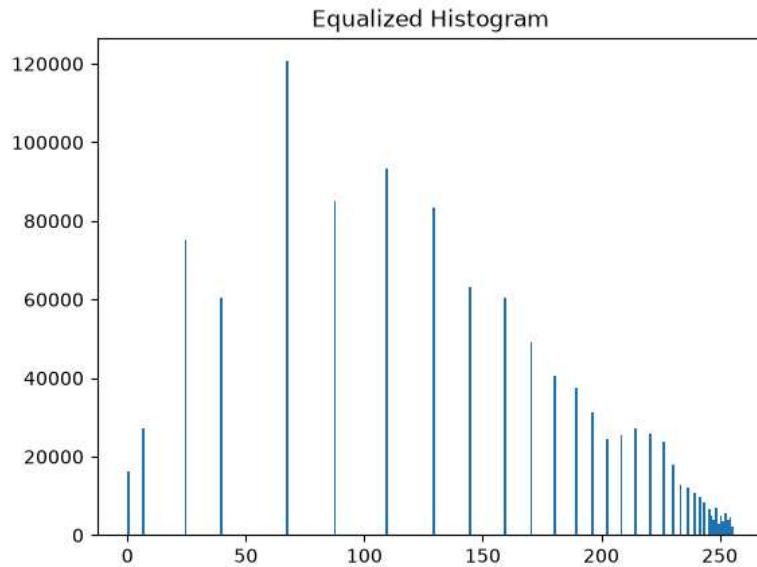
```
plt.hist(img.ravel(),256,range = [0, 256]);  
plt.title('Original Image')  
plt.show()
```



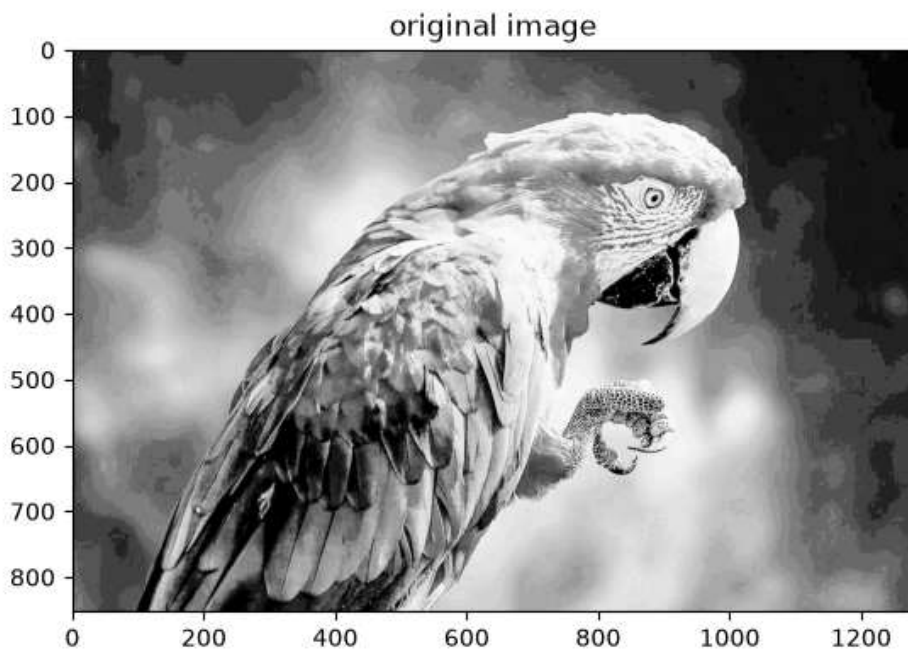
```
img_eq = cv2.equalizeHist(img)  
plt.hist(img_eq.ravel(), 256, range = [0, 256]);  
plt.title('Equalized Histogram')
```



```
[11]: Text(0.5, 1.0, 'Equalized Histogram')
```



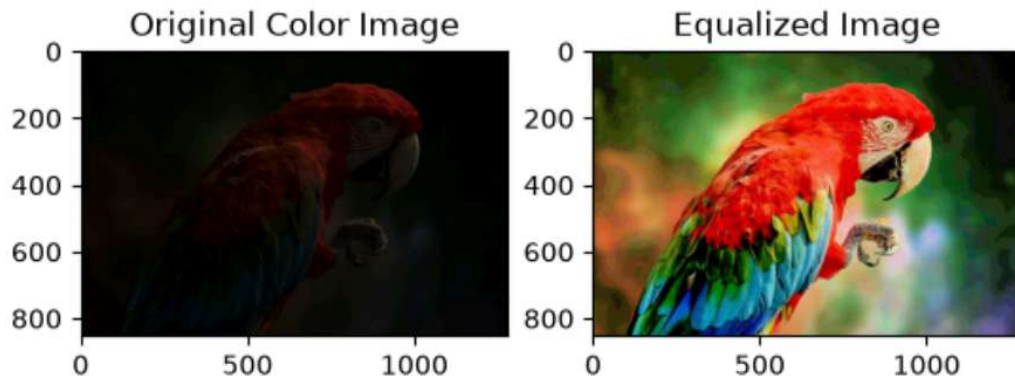
```
plt.imshow(img_eq, cmap='gray')  
plt.title('original image')  
plt.show()
```



```
img = cv2.imread('parrot.jpg', cv2.IMREAD_COLOR)  
img_hsv = cv2.cvtColor(img, cv2.COLOR_BGR2HSV)  
img_hsv[:, :, 2] = cv2.equalizeHist(img_hsv[:, :, 2])  
img_eq = cv2.cvtColor(img_hsv, cv2.COLOR_HSV2BGR)  
plt.subplot(121); plt.imshow(img[:, :, ::-1]); plt.title('Original Color Image')  
plt.subplot(122); plt.imshow(img_eq[:, :, ::-1]); plt.title('Equalized Image')
```

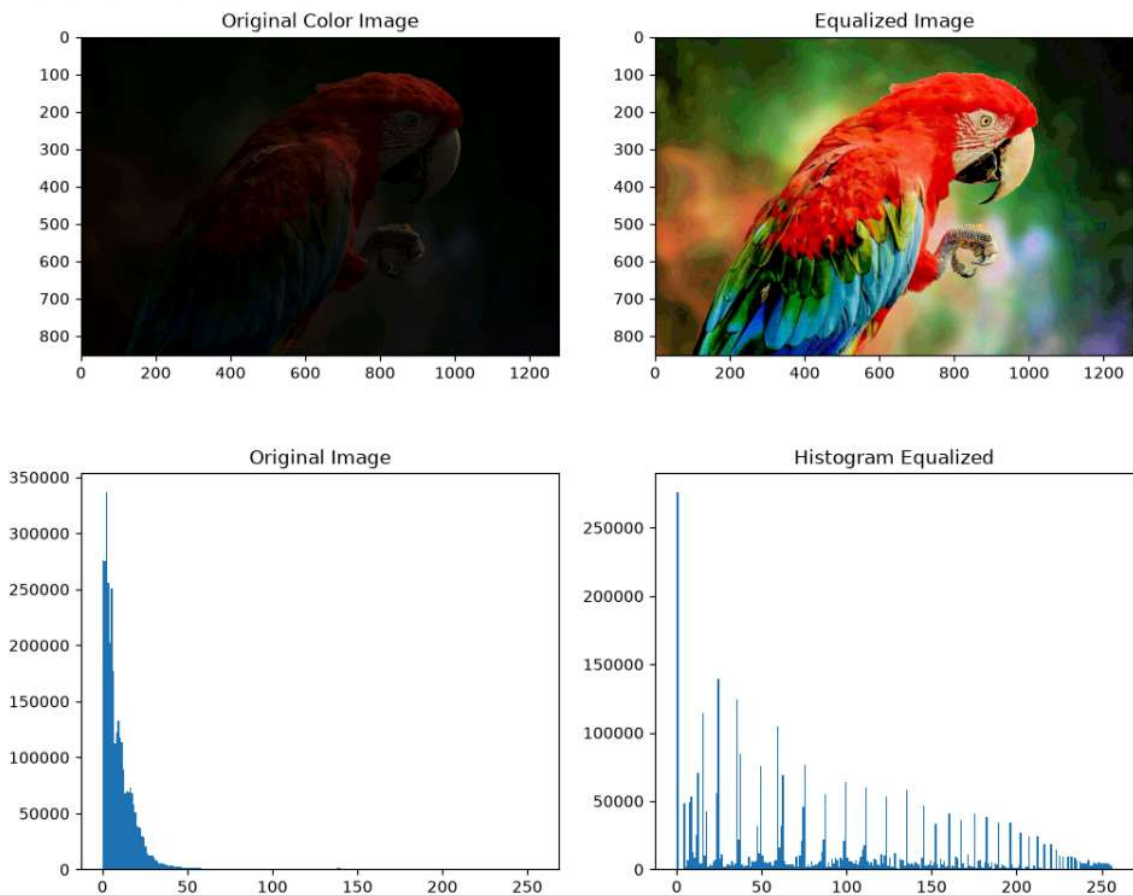


```
[13]: Text(0.5, 1.0, 'Equalized Image')
```



```
plt.figure(figsize = [12,10])  
plt.subplot(221); plt.imshow(img[:, :, ::-1]); plt.title('Original Color Image')  
plt.subplot(222); plt.imshow(img_eq[:, :, ::-1]); plt.title('Equalized Image')  
plt.subplot(223); plt.hist(img.ravel(),256,range = [0, 256]); plt.title('Original Image')  
plt.subplot(224); plt.hist(img_eq.ravel(),256,range = [0, 256]); plt.title('Histogram  
Equalized')
```

```
[15]: Text(0.5, 1.0, 'Histogram Equalized')
```



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